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Cambridge IGCSE Combined and Co-ordinated Sciences

B4 Enzymes

Condensed Version — Key Concepts and High-Probability Exam Questions

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Cambridge IGCSE Biology · The most important concepts and exam questions for your test

Must-Know Concepts

Enzymes as Biological Catalysts

- **Catalyst:** a substance that **increases the rate of a chemical reaction** and **is not changed by the reaction**. You need both parts.
- **Enzymes** are **proteins** (made of **amino acids**) that are involved in all **metabolic reactions**, where they function as **biological catalysts**.
- Without enzymes, metabolic reactions would be very slow or would not happen at all: enzymes make them **fast enough to sustain life**.
- Names ending in **-ase** are enzymes (**carbohydrases**, **proteases**, **lipases**); names ending in **-ose** are sugars.

How Enzymes Work

1. The **substrate** has a shape that is **complementary** to the enzyme's **active site**.
2. The substrate fits into the active site and binds to the enzyme.
3. The enzyme changes the substrate into **product(s)**, for example by splitting one molecule into two, or joining two into one.
4. The products break away. The enzyme is **unchanged** and ready to bind another substrate molecule.

Enzyme–Substrate Complex and Specificity (Supplement)

- When the substrate binds to the active site it forms a short-lived **enzyme–substrate complex**: **enzyme + substrate → enzyme–substrate complex → enzyme + product(s)**.
- **Specificity:** each enzyme acts on only one type of substrate, because only a substrate with a shape **complementary** to the active site can fit and bind (the **lock and key model**). For example, amylase breaks down starch but not proteins or fats.

Effects of Temperature and pH

- **Temperature graph:** the rate is zero at 0 °C, rises slowly at first, then steeply from about 20 °C to 35 °C, to a peak at the **optimum, about 37–38 °C**, then falls much more steeply to **zero at about 58–60 °C**.

Condition	Temperature	pH
Optimum	About 37 °C (human enzymes); up to 80 °C in some hot-spring bacteria	About pH 7 (most enzymes); about pH 2 (stomach protease)
Below the optimum	Slow (little kinetic energy) but NOT denatured; works again when warmed	Shape changes, so the active site fits less well; denatured if far below
Above the optimum	Starts to lose its shape; many enzymes are completely denatured above about 60 °C	Shape changes, so the active site fits less well; denatured if far above

Explaining the Effects (Supplement)

1. **Rising temperature:** enzyme and substrate molecules gain **kinetic energy**, move faster and collide more frequently, so there are more **effective collisions** and more enzyme–substrate complexes form per second.

2. **At the optimum temperature:** there are the most effective collisions while the active site still has its correct shape.
3. **Above the optimum:** the kinetic energy begins to shake the enzyme apart, so its active site is no longer **complementary** to the substrate and fewer enzyme–substrate complexes form. At about 60 °C it is completely **denatured**.

- **pH** is explained by **shape**, never kinetic energy: away from the optimum the active site changes shape and fits the substrate less well; far from it the enzyme is denatured and cannot bind the substrate.

Key Terms

Term	Definition
Catalyst	A substance that increases the rate of a chemical reaction and is not changed by the reaction
Enzymes	Proteins that are involved in all metabolic reactions, where they function as biological catalysts
Substrate	The substance that an enzyme causes to react (e.g. starch for amylase)
Active site	The part of an enzyme molecule to which the substrate temporarily binds
Complementary	With a perfect mirror-image shape, so the active site and its substrate fit together
Enzyme–substrate complex (Supplement)	The short-lived structure formed as the substrate binds temporarily to the active site of an enzyme
Specificity (Supplement)	Of enzymes, only able to act on a particular (specific) substrate
Optimum temperature	The temperature at which an enzyme works fastest
Optimum pH	The pH at which an enzyme works fastest
Denaturation	Damage to an enzyme's shape (including its active site) caused by high temperature or an unsuitable pH; the enzyme can no longer bind its substrate or catalyse its reaction
Effective collision (Supplement)	A collision in which a substrate molecule enters an enzyme's active site, forms an enzyme–substrate complex and is converted to product
Buffer solution	A liquid that has a known pH, and keeps that pH steady all the time, even if chemical reactions take place
Anomalous result	A result that does not fit the pattern of the other results; it is left out when calculating a mean

Key Reactions, Numbers and Methods

Reactions Catalysed by Enzymes

Enzyme	Type	Reaction catalysed
Amylase	Carbohydrase	starch → maltose
Maltase	Carbohydrase	maltose → glucose
Lactase	Carbohydrase	lactose → smaller sugars (glucose + galactose)

Enzyme	Type	Reaction catalysed
Protease	Protease	protein → amino acids
Lipase	Lipase	lipid (fats and oils) → fatty acids + glycerol
Catalase	Works inside all cells	hydrogen peroxide → water + oxygen

Rate, Time and Means

- **rate = 1 ÷ time taken**, so a shorter time means a faster rate and the **shortest time is at the optimum**. Graphs of time against temperature or pH are roughly **U-shaped**.
- Example: a time of 4 min gives a rate of 1 ÷ 4 = 0.25 per min; a time of 8 min gives 1 ÷ 8 = 0.125 per min; so the first reaction is twice as fast.
- **Mean** = (1st + 2nd + 3rd) ÷ 3, leaving out any anomalous result.

Tests and Indicators

- **Iodine solution:** turns **blue-black** if starch is present; stays **brown** if there is no starch.
- **Thymolphthalein: blue** in alkaline conditions; **loses its colour** when the pH becomes less alkaline (e.g. when lipase makes fatty acids).
- **Oxygen** from catalase relights a glowing splint. Boiled potato gives no froth, because its catalase is denatured.

Investigating Temperature and pH

- Change the **temperature** with water-baths and the **pH** with **buffer solutions**, and measure the **time** taken for a change.
- Keep the same: volume and concentration of the enzyme and substrate solutions, volume of buffer, and temperature (when testing pH). Be specific, never just "the amount".
- Improve: repeat and calculate means, test more values closer together around the optimum, use a thermostatically controlled water-bath and use fresh solutions.

High-Probability Exam Questions

Question 1

Explain why enzymes are called biological catalysts. (4 marks)

Answer:

- They increase the rate of chemical reactions.
- They are not changed by the reactions, so they can be used again and again.
- They are proteins made by living cells, and they catalyse the metabolic reactions inside living organisms.
- Without enzymes these reactions would be far too slow to sustain life.

Question 2

A diagram of an enzyme-controlled reaction shows: an enzyme with a dent (**Q**) beside a round molecule (**P**); the round molecule fitted into the dent (the pair labelled **R**); the enzyme with its dent empty, beside the round molecule split in two (**S**). Identify P, Q, R and S. (4 marks)

Answer:

- **P** = substrate
- **Q** = active site
- **R** = enzyme–substrate complex
- **S** = products

Question 3

(Supplement) Explain why enzymes are specific. (4 marks)

Answer:

- The active site of an enzyme has a very specific shape.
- Only a substrate molecule with a complementary shape can fit into the active site.
- The substrate binds to the active site to form an enzyme–substrate complex.
- Molecules of other substances have different shapes, so they do not fit and cannot be changed by that enzyme (like a key that does not fit a lock).

Question 4

(Supplement) On a graph of rate of reaction against pH, the curve for most enzymes is at zero below about pH 5.6, rises very steeply to a peak at pH 7, then falls more gradually to zero at about pH 10. Explain the shape of this curve. (4 marks)

Answer:

- At the optimum (about pH 7) the active site is the best fit for (complementary to) the substrate, so many enzyme–substrate complexes form and the rate is highest.
- As the pH moves away from 7, the shape of the enzyme, including its active site, begins to change.
- The active site becomes less complementary, so fewer substrate molecules fit and bind, and the rate falls.
- Below about pH 5.6 or above about pH 10 the enzyme has lost its shape completely (it is denatured), so the substrate cannot bind and the rate is zero.

Question 5

Amylase is an enzyme that breaks down starch to maltose.

(a) A student tested starch solution with iodine solution (result: **blue-black**), added amylase solution, then tested a sample again after five minutes (result: **brown**). Explain the student's results. [3]

(b) He then put the same volume of starch solution into five tubes, added a different buffer solution and then amylase solution to each, and measured the time for all the starch to disappear.

pH	4	6	7	9	11
Time for starch to disappear / min	did not disappear	8	4	14	did not disappear

(i) Use the results to describe the effect of pH on the activity of amylase. [3]

(ii) State three other variables that the student should have kept the same. [3] (9 marks)

Answer:

- **(a)** Blue-black at the start shows that starch was present.
- **(a)** Amylase broke down (digested) the starch into maltose.
- **(a)** Brown after five minutes shows that no starch was left.
- **(b)(i)** Amylase is most active at pH 7, its optimum: the starch disappeared in the shortest time (4 min).
- **(b)(i)** Activity decreases as the pH moves away from 7 in either direction: 8 min at pH 6 and 14 min at pH 9.
- **(b)(i)** At pH 4 and pH 11 the starch did not disappear, so the amylase was not working (it was denatured).
- **(b)(ii)** Temperature (e.g. all the tubes in the same water-bath).
- **(b)(ii)** Volume (or concentration) of amylase solution.
- **(b)(ii)** Concentration of starch solution.

Question 6

A graph shows rate of activity against temperature for two enzymes. **Enzyme A** rises from zero at 10 °C to a peak at 40 °C, then falls steeply to zero at about 58 °C. **Enzyme B** rises from zero at 40 °C (about 70% of its maximum at 60 °C) to a peak at 70 °C, then falls to zero at about 92 °C. Both peaks are about the same height.

(a) Compare the effect of temperature on enzyme A and enzyme B. [5]

(b) (Supplement) Explain the effect of temperature on enzyme B from **(i)** 40 to 60 °C [3] and **(ii)** 70 to 90 °C. [4] (12 marks)

Answer:

- **(a)** For both enzymes, activity increases with temperature up to an optimum, then decreases.
- **(a)** Enzyme A has an optimum of 40 °C, whereas enzyme B's is 70 °C, 30 °C higher.
- **(a)** Enzyme A starts to work at 10 °C, but enzyme B shows no activity until 40 °C.
- **(a)** Enzyme A is completely denatured at about 58 °C, whereas enzyme B keeps working until about 92 °C.
- **(a)** Both enzymes reach about the same maximum rate of activity.
- **(b)(i)** The activity of enzyme B increases from zero at 40 °C to about 70% of its maximum at 60 °C.
- **(b)(i)** The enzyme and substrate molecules gain kinetic energy and move faster.
- **(b)(i)** They collide more frequently, so there are more effective collisions and more enzyme–substrate complexes form per second.
- **(b)(ii)** The activity of enzyme B decreases from its maximum at 70 °C to a low level by 90 °C.
- **(b)(ii)** The high kinetic energy makes the enzyme vibrate and begin to shake apart, so its active site changes shape.
- **(b)(ii)** The active site is no longer complementary to the substrate, so fewer enzyme–substrate complexes form.
- **(b)(ii)** The enzyme is denatured; by about 92 °C it is completely denatured.

Question 7

A filter paper square dipped in catalase solution is pushed to the bottom of a beaker of hydrogen peroxide and buffer solution. The time for it to rise to the top is recorded.

(a) Write the word equation for the reaction. [1]

(b) Explain why the paper rises to the top. [3]

(c) At one pH, three tries took 12 s, 13 s and 31 s. Identify the anomalous result and calculate the mean. [2] (6 marks)

Answer:

- **(a)** hydrogen peroxide → water + oxygen
- **(b)** Catalase on the paper breaks down the hydrogen peroxide, producing oxygen gas.
- **(b)** Oxygen bubbles collect on the paper and make it buoyant, so it floats up.
- **(b)** The faster the catalase works, the sooner the paper rises, so a shorter time means a faster rate.
- **(c)** 31 s is anomalous (it does not fit the pattern of the other two), so leave it out.
- **(c)** Mean = (12 + 13) ÷ 2 = 12.5 s.

Question 8

Milk (which contains lipids) is mixed with sodium carbonate solution and thymolphthalein indicator, then lipase is added. The time for the blue colour to disappear is measured at different temperatures.

(a) Explain why the liquid was blue at the start. [2]

(b) Explain why the liquid in some tubes became colourless. [4] (6 marks)

Answer:

- **(a)** Sodium carbonate solution is alkaline.
- **(a)** Thymolphthalein is blue in alkaline conditions.
- **(b)** Lipase broke down the lipids in the milk into fatty acids and glycerol.
- **(b)** The fatty acids lowered the pH, so the mixture became less alkaline.
- **(b)** Thymolphthalein loses its colour when the pH becomes less alkaline.
- **(b)** This only happened where the lipase was working; in very hot tubes it was denatured, so the liquid stayed blue.

Common Mistakes to Avoid

- **Enzymes are denatured, never "killed":** they are molecules, not living things.
- **Low temperatures do not denature enzymes:** cold enzymes are just slow, and work again when warmed.
- **The active site is complementary to the substrate**, not "the same shape".
- **Explain pH effects by a change in shape:** kinetic energy and collisions are for temperature only.
- **The shortest time means the fastest rate** (rate = 1 ÷ time).
- **"Describe" a graph with figures and units** (e.g. "a maximum at about 37 °C"); "explain" means give the reasons.
- **Iodine stays brown when there is no starch:** it never "turns clear".
- **Spell carefully:** maltose (a sugar) and maltase (an enzyme); lipid (substrate) and lipase (enzyme).

Last-Minute Checklist

- I can define a catalyst and explain why enzymes are biological catalysts needed to sustain life.
- I can write the word equations for amylase, maltase, protease, lipase, catalase and lactase.
- I can describe how an enzyme works using active site, substrate, complementary shape and product.
- I can use the term enzyme–substrate complex and explain why enzymes are specific. (Supplement)
- I can state the optimums: about 37 °C for human enzymes, about pH 7 for most enzymes, about pH 2 for the stomach protease.
- I can describe temperature and pH graphs with figures, units and the words optimum and denatured.
- I can explain temperature effects using kinetic energy, effective collisions, complementary shape and denaturation. (Supplement)
- I can explain pH effects using complementary shape and denaturation. (Supplement)
- I can describe a temperature or pH investigation and name its variables.
- I can turn times into rates, find a mean without the anomalous result, and read iodine and thymolphthalein colours.